CAMBRIDGE
International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

8720963317

MATHEMATICS **9709/43**

Paper 4 Mechanics **May/June 2026**

1 hour 15 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use $10\,\text{m}\,\text{s}^{-2}$.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages.

© Cambridge University Press & Assessment 2026

1 A particle is projected vertically upwards with speed $u\,\text{m}\,\text{s}^{-1}$ from a point on horizontal ground. The height of the particle above the ground after 4 seconds is equal to its height above the ground after 6 seconds.

(a) Find the value of u. [2]

(b) Find the greatest height of the particle above the ground. [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

2 A particle P moves in a straight line. The velocity of P at time t s is v m s^{-1}, where

$$v = 2t^2 - 6t - 3.$$

(a) Find the time at which the acceleration of P is zero. [2]

.....

.....

.....

.....

.....

.....

.....

(b) Given that P is not at rest in the interval $0 \leqslant t \leqslant 3$, find the distance travelled by P in this interval. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

3 A block of mass m kg is pulled up a rough plane inclined at an angle of 30° to the horizontal by a constant force of magnitude 20 N. This force acts parallel to a line of greatest slope of the plane. The acceleration of the block is $2\,\text{m}\,\text{s}^{-2}$. The coefficient of friction between the block and the plane is 0.4.

Find the value of m. [4]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

4 A train is travelling along a straight track with constant acceleration $a\,\text{m s}^{-2}$. It passes through points A, B and C on the track in that order. The distances AB and BC are 280 m and 170 m respectively. The speed of the train at B is $32\,\text{m s}^{-1}$. The time it takes the train to travel from A to B is $2t$ s and from B to C is t s.

Find the value of a and the value of t. [5]

5 A lorry of mass 13 500 kg is towing a trailer of mass 2500 kg along a straight horizontal road. The lorry and the trailer are connected by a light rigid tow-bar which is parallel to the road.

(a) At a particular instant, the power of the lorry's engine is 80 kW and the resistance forces on the lorry and trailer are 2400 N and 400 N respectively. At this instant, the speed of the lorry is $25\,\text{m}\,\text{s}^{-1}$.

Find the tension in the tow-bar. [5]

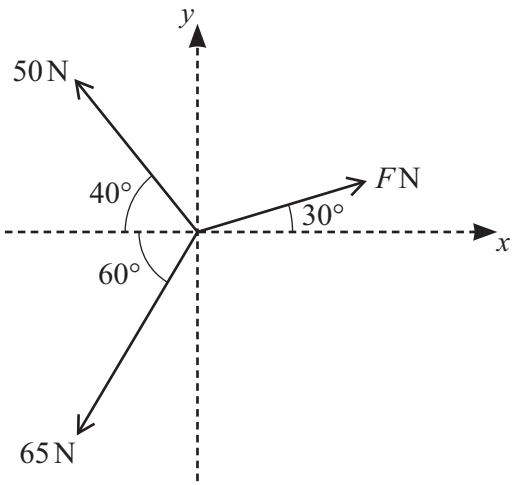
DO NOT WRITE IN THIS MARGIN

The lorry and trailer now descend a straight hill inclined at an angle of 5° to the horizontal. As the lorry and trailer descend from a point A to a point B on the hill, the speed of the lorry increases from $27\,\text{m s}^{-1}$ to $30\,\text{m s}^{-1}$. The lorry's engine is no longer working at 80 kW and instead the work done by the driving force of the lorry as it travels from A to B is 55 kJ. The resistance forces on the lorry and trailer are now 2500 N and 500 N respectively.

(b) Find the distance AB. [5]

DO NOT WRITE IN THIS MARGIN

6



Three coplanar forces of magnitudes F N, 65 N and 50 N act at a point in the directions shown in the diagram.

(a) It is initially given that $F = 40$.

Find the magnitude and direction of the resultant force. [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

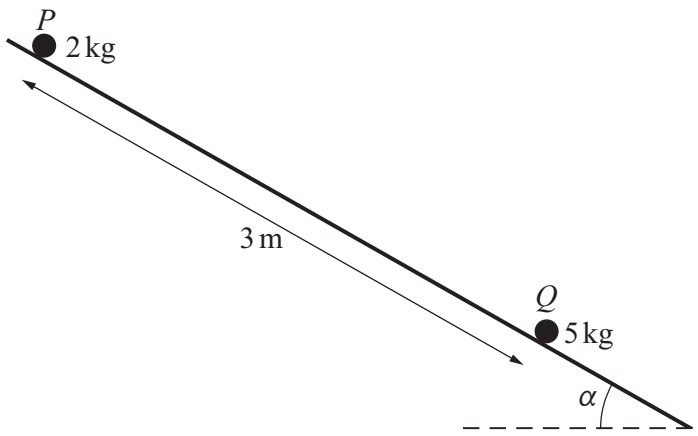
© Cambridge University Press & Assessment 2026

(b) It is given instead that the value of F is such that the component of the resultant force in the y-direction is zero.

Find the magnitude and direction of the resultant force. [3]

DO NOT WRITE IN THIS MARGIN

7



Particles P and Q have masses 2 kg and 5 kg respectively. The particles are initially held at rest 3 m apart on the same line of greatest slope of a rough plane. The plane is inclined at an angle α to the horizontal, where $\sin\alpha = 0.28$ (see diagram). The coefficient of friction between P and the plane and the coefficient of friction between Q and the plane are both 0.2.

Particle P is projected directly towards Q with speed $8\,\text{m s}^{-1}$. At the same instant, Q is projected directly towards P with speed $u\,\text{m s}^{-1}$. The particles collide 0.25 s after they are projected.

(a) Show that the acceleration of P is $0.88\,\text{m s}^{-2}$, and find the value of u. [8]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

© Cambridge University Press & Assessment 2026

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) When the particles collide, P rebounds up the plane, and Q rebounds down the plane with speed $0.5\,\text{m}\,\text{s}^{-1}$.

Find the distance that P travels from the point of collision, until it comes to instantaneous rest. [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

© Cambridge University Press & Assessment 2026